

1. Symbolic algebra in Factor

A tour of `math.symbolic` and its companion vocabularies, as forty-odd worked problems. Every Factor block here is run by `misc/symbolic-tutorial/check.py`, which compares what Factor prints against the `! =>` lines, so the outputs in this document are the real ones.

Expressions are written in postfix inside `symbolic[ ... ]`: numbers are numbers, `+` `-` `*` `/` `^` `neg` `sqrt` `sin` `cos` `tan` `exp` `log` and the inverse, hyperbolic and gamma functions build expressions, `pi`, `e` and `inf` are constants, `?x` is a pattern variable, and any other token is a variable. `expr.` prints an expression in ordinary notation, while `.` prints the underlying value.

2. Expressions and simplification

2.1. Like terms and powers

$$x + x, \quad x \cdot x \cdot x, \quad \frac{x^2}{x}$$

Sums and products are simplified as they are built.

```
symbolic[ x x + ] expr.  
! => 2*x  
symbolic[ x x * x * ] expr.  
! => x^3  
symbolic[ x 2 ^ x / ] expr.  
! => x  
symbolic[ x x - ] .  
! => 0
```

2.2. Exact arithmetic

Rationals stay exact, and a number multiplying a sum is distributed.

$$\frac{1}{2} + \frac{1}{3} = \frac{5}{6}, \quad 2(x + 1) = 2x + 2$$

```
symbolic[ 1 2 / 1 3 / + ] .  
! => 5/6  
symbolic[ 2 x 1 + * ] expr.  
! => 2*x + 2  
symbolic[ e e 1 - - ] .  
! => 1
```

2.3. Expanding

$$(x + 1)^2, \quad (x + y)(x - y)$$

```
symbolic[ x 1 + 2 ^ ] expand expr.  
! => x^2 + 2*x + 1  
symbolic[ x y + x y - * ] expand expr.  
! => x^2 - y^2
```

2.4. Printing and reading back

An expression `prettyprints` as a literal that reads back in, while `expr.` gives ordinary notation.

```

symbolic[ 3 x sin * x + ] .
! => symbolic[ x 3 x sin * + ]
symbolic[ 3 x sin * x + ] expr.
! => x + 3*sin(x)
symbolic[ 1 x / ] expr.
! => 1/x
symbolic[ 2 3 x * / ] expr.
! => 2/(3*x)

```

2.5. Substituting and evaluating

$$x^2 + 1 \text{ at } x = 2, \quad \sin\left(\frac{\pi}{4}\right)\sqrt{2}$$

```

symbolic[ x 2 ^ 1 + ] { { T{ sym f "x" } 2 } } subs .
! => 5
symbolic[ x x sin + ] { { T{ sym f "x" } T{ sym f "y" } } } subs expr.
! => y + sin(y)
symbolic[ pi 4 / sin 2 sqrt * ] evalf .
! => 1.0

```

2.6. Exact angles

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}, \quad \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}, \quad \tan\left(\frac{\pi}{3}\right) = \sqrt{3}$$

The inverse functions give the same angles back.

```

symbolic[ pi 6 / sin ] .
! => 1/2
symbolic[ pi 6 / cos ] expr.
! => sqrt(3)/2
symbolic[ pi 3 / tan ] expr.
! => sqrt(3)
symbolic[ 1 2 / asin ] expr.
! => pi/6
symbolic[ 3 sqrt atan ] expr.
! => pi/3

```

3. Differentiation

3.1. The product rule

$$\frac{d}{dx}(x \sin x) = x \cos x + \sin x$$

```

symbolic[ x x sin * ] symbolic[ x ] differentiate expr.
! => x*cos(x) + sin(x)

```

3.2. The chain rule

$$\frac{d}{dx}e^{x^2} = 2xe^{x^2}$$

```
symbolic[ x 2 ^ exp ] symbolic[ x ] differentiate expr.
! => 2*x*exp(x^2)
symbolic[ 3 x * 1 + sin ] symbolic[ x ] differentiate expr.
! => 3*cos(3*x + 1)
```

3.3. Quotients and powers

$$\frac{d}{dx} \frac{x}{x^2 + 1}, \quad \frac{d}{dx} x^x$$

```
symbolic[ x x 2 ^ 1 + / ] symbolic[ x ] differentiate expand expr.
! => -2*x^2/(x^2 + 1)^2 + 1/(x^2 + 1)
symbolic[ x x ^ ] symbolic[ x ] differentiate expr.
! => x^x*(log(x) + 1)
```

3.4. Inverse and hyperbolic functions

$$\frac{d}{dx} \arctan x = \frac{1}{x^2 + 1}, \quad \frac{d}{dx} \tanh x = \frac{1}{\cosh^2 x}$$

```
symbolic[ x atan ] symbolic[ x ] differentiate expr.
! => 1/(x^2 + 1)
symbolic[ x asin ] symbolic[ x ] differentiate expr.
! => 1/sqrt(-x^2 + 1)
symbolic[ x tanh ] symbolic[ x ] differentiate expr.
! => 1/cosh(x)^2
```

3.5. Gradients and Jacobians

For the polar map $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$ the Jacobian determinant is the r of $dx dy = r dr d\theta$.

```
CONSTANT: polar { T{ sym f "r" } T{ sym f "theta" } }

symbolic[ r theta cos * ] symbolic[ r theta sin * ] 2array polar jacobian
[ [ expr. ] each ] each
! => cos(theta)
! => -r*sin(theta)
! => sin(theta)
! => r*cos(theta)
```

```
CONSTANT: polar { T{ sym f "r" } T{ sym f "theta" } }

:: det2 ( matrix -- expr )
  matrix first first2 := ( a b )
  matrix second first2 := ( c d )
  a d s* b c s* s- ;

symbolic[ r theta cos * ] symbolic[ r theta sin * ] 2array polar jacobian
det2 pythagorean-rules rewrite expr.
! => r
```

3.6. Classifying a critical point

$$f(x, y) = x^3 - 3xy + y^3$$

The gradient vanishes at (1, 1) and at the origin. The Hessian determinant settles which is which.

```
CONSTANT: vars { T{ sym f "x" } T{ sym f "y" } }
CONSTANT: f symbolic[ x 3 ^ 3 x * y * - y 3 ^ + ]

f vars gradient [ expr. ] each
! => 3*x^2 - 3*y
! => 3*y^2 - 3*x
f vars hessian [ [ expr. ] each ] each
! => 6*x
! => -3
! => -3
! => 6*y
```

With $f_{xx} = 6 > 0$ and determinant $27 > 0$ at (1, 1) it is a local minimum; at the origin the determinant is negative, so that is a saddle.

```
CONSTANT: vars { T{ sym f "x" } T{ sym f "y" } }
CONSTANT: f symbolic[ x 3 ^ 3 x * y * - y 3 ^ + ]

:: det2 ( matrix -- expr )
  matrix first first2 :> ( a b )
  matrix second first2 :> ( c d )
  a d s* b c s* s- ;

f vars hessian det2 expr.
! => 36*x*y - 9
f vars hessian det2 { { T{ sym f "x" } 1 } { T{ sym f "y" } 1 } } subs .
! => 27
f vars hessian det2 { { T{ sym f "x" } 0 } { T{ sym f "y" } 0 } } subs .
! => -9
```

3.7. Derivatives left unevaluated

D writes a derivative down without computing it; doit computes it.

```
symbolic[ x 3 ^ x D ] expr.
! => D(x^3, x)
symbolic[ x 3 ^ x D ] doit expr.
! => 3*x^2
```

4. Antiderivatives

4.1. Powers, trigonometry and exponentials

$$\int x^2 dx = \frac{x^3}{3}, \quad \int \frac{1}{x} dx = \log x$$

```
symbolic[ x 2 ^ ] symbolic[ x ] integrate expr.
! => x^3/3
symbolic[ 1 x / ] symbolic[ x ] integrate expr.
! => log(x)
symbolic[ x exp ] symbolic[ x ] integrate expr.
! => exp(x)
```

4.2. Linear arguments

$$\int \sin 2x \, dx = -\frac{\cos(2x)}{2}, \quad \int \frac{1}{2x+1} \, dx$$

```
symbolic[ 2 x * sin ] symbolic[ x ] integrate expr.  
! => -cos(2*x)/2  
symbolic[ 1 2 x * 1 + / ] symbolic[ x ] integrate expr.  
! => log(2*x + 1)/2  
symbolic[ 3 x * 1 + exp ] symbolic[ x ] integrate expr.  
! => exp(3*x + 1)/3
```

4.3. An arc tangent and a logarithm

$$\int \frac{1}{x^2+1} \, dx = \arctan x, \quad \int \tan x \, dx$$

```
symbolic[ x 2 ^ 1 + -1 ^ ] symbolic[ x ] integrate expr.  
! => atan(x)  
symbolic[ x tan ] symbolic[ x ] integrate expr.  
! => -log(cos(x))
```

4.4. Squares of sine and cosine

$$\int \sin^2 x \, dx = \frac{x}{2} - \frac{\sin 2x}{4}$$

```
symbolic[ x sin 2 ^ ] symbolic[ x ] integrate expr.  
! => x/2 - sin(2*x)/4  
symbolic[ x sin x cos * ] symbolic[ x ] integrate expr.  
! => -cos(x)^2/2
```

4.5. When the derivative divides

$$\int 2xe^{x^2} \, dx = e^{x^2}$$

This one is found automatically: the rest of the product is a constant times the derivative of the inner expression.

```
symbolic[ 2 x * x 2 ^ exp * ] symbolic[ x ] integrate expr.  
! => exp(x^2)  
symbolic[ x x 2 ^ 1 + / ] symbolic[ x ] integrate expr.  
! => log(x^2 + 1)/2
```

4.6. No elementary antiderivative

The integral comes back unevaluated rather than wrong.

```
symbolic[ x 2 ^ neg exp ] symbolic[ x ] integrate expr.  
! => integrate(exp(-x^2), x)  
symbolic[ x sin x / ] symbolic[ x ] integrate expr.  
! => integrate(sin(x)/x, x)
```

5. Integration by hand

5.1. By parts, choosing u

$$\int x e^x dx = x e^x - \int e^x dx$$

by-parts-u leaves the remaining integral unevaluated, so the step is visible; doit finishes it.

```
symbolic[ x x exp * ] symbolic[ x ] symbolic[ x ] by-parts-u expr.  
! => x*exp(x) - integrate(exp(x), x)  
symbolic[ x x exp * ] symbolic[ x ] symbolic[ x ] by-parts-u doit expr.  
! => x*exp(x) - exp(x)
```

5.2. By parts, choosing dv

$$\int x \log x dx, \quad dv = x dx$$

Choosing dv instead fixes $u = \log x$, whose derivative is simple.

```
symbolic[ x x log * ] symbolic[ x ] symbolic[ x ] by-parts-dv expr.  
! => x^2*log(x)/2 - integrate(x/2, x)  
symbolic[ x x log * ] symbolic[ x ] symbolic[ x ] by-parts-dv doit expr.  
! => x^2*log(x)/2 - x^2/4
```

5.3. The $dv = 1$ trick

$$\int \log x dx = x \log x - \int 1 dx$$

For a single logarithm there is nothing else to be dv , so take $dv = dx$.

```
symbolic[ x log ] symbolic[ x ] 1 by-parts-dv expr.  
! => x*log(x) - integrate(1, x)  
symbolic[ x log ] symbolic[ x ] 1 by-parts-dv doit expr.  
! => -x + x*log(x)
```

5.4. Repeated parts

$$\int x^2 \sin x dx$$

Each application lowers the power of x by one.

```
symbolic[ x 2 ^ x sin * ] symbolic[ x ] symbolic[ x 2 ^ ] by-parts-u expr.  
! => -x^2*cos(x) - integrate(-2*x*cos(x), x)  
symbolic[ x 2 ^ x sin * ] symbolic[ x ] integrate expr.  
! => -x^2*cos(x) + 2*x*sin(x) + 2*cos(x)
```

5.5. A substitution you choose

$$\int 2x e^{x^2} dx, \quad u = x^2$$

```

symbolic[ 2 x * x 2 ^ exp * ] symbolic[ x ] symbolic[ x 2 ^ ]
by-substitution expr.
! => exp(x^2)
symbolic[ x sin x cos * ] symbolic[ x ] symbolic[ x sin ]
by-substitution expr.
! => sin(x)^2/2

```

The integrand has to mention the substituted expression after dividing by u' : $\sin x / \cos x$ fits $u = \cos x$, while the equal expression $\tan x$ does not.

```

symbolic[ x sin x cos / ] symbolic[ x ] symbolic[ x cos ]
by-substitution expr.
! => -log(cos(x))
symbolic[ x tan ] symbolic[ x ] symbolic[ x cos ] by-substitution expr.
! => integrate(tan(x), x)

```

6. Definite, improper and special integrals

6.1. Definite integrals

$$\int_0^{\pi} \sin x \, dx = 2, \quad \int_0^1 x^2 \, dx = \frac{1}{3}$$

```

symbolic[ x sin ] symbolic[ x ] 0 symbolic[ pi ] definite-integrate .
! => 2
symbolic[ x 2 ^ ] symbolic[ x ] 0 1 definite-integrate .
! => 1/3
symbolic[ x exp ] symbolic[ x ] 0 1 definite-integrate expr.
! => e - 1

```

6.2. Substituting the limits

A definite substitution maps the limits to $u(a)$ and $u(b)$, so there is nothing to substitute back.

$$\int_0^1 2xe^{x^2} \, dx = e - 1$$

```

symbolic[ 2 x * x 2 ^ exp * ] symbolic[ x ] 0 1 symbolic[ x 2 ^ ]
definite-by-substitution expr.
! => e - 1
symbolic[ x sin x cos * ] symbolic[ x ] 0 symbolic[ pi 2 / ]
symbolic[ x sin ] definite-by-substitution .
! => 1/2

```

6.3. To infinity

$$\int_0^{\infty} e^{-x} \, dx = 1, \quad \int_1^{\infty} \frac{1}{x^2} \, dx = 1$$

```

symbolic[ x neg exp ] symbolic[ x ] 0 symbolic[ inf ] definite-integrate .
! => 1
symbolic[ x -2 ^ ] symbolic[ x ] 1 symbolic[ inf ] definite-integrate .

```

```
! => 1
symbolic[ x 2 ^ 1 + -1 ^ ] symbolic[ x ] 0 symbolic[ inf ]
definite-integrate expr.
! => pi/2
```

6.4. Divergence, and a singular endpoint

$$\int_1^{\infty} \frac{1}{x} dx = \infty, \quad \int_0^1 \log x dx = -1$$

The second integrand is unbounded at 0, and the endpoint is evaluated as a limit.

```
symbolic[ 1 x / ] symbolic[ x ] 1 symbolic[ inf ] definite-integrate expr.
! => inf
symbolic[ x log ] symbolic[ x ] 0 1 definite-integrate .
! => -1
```

6.5. The Gaussian integral

$$\int_{-\infty}^{\infty} e^{-(x-5)^2/9} dx = 3\sqrt{\pi}$$

There is no elementary antiderivative, so this is a rule of its own.

```
symbolic[ x 2 ^ neg exp ] symbolic[ x ] symbolic[ inf neg ] symbolic[ inf ]
definite-integrate expr.
! => sqrt(pi)
symbolic[ x 5 - 2 ^ 9 / neg exp ] symbolic[ x ]
symbolic[ inf neg ] symbolic[ inf ] definite-integrate expr.
! => 3*sqrt(pi)
```

6.6. The gamma function

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{\Gamma(n+1)}{a^{n+1}}$$

```
symbolic[ x 2 ^ x neg exp * ] symbolic[ x ] 0 symbolic[ inf ]
definite-integrate .
! => 2
symbolic[ x 2 x * neg exp * ] symbolic[ x ] 0 symbolic[ inf ]
definite-integrate .
! => 1/4
symbolic[ 5 gamma ] .
! => 24
symbolic[ 3 2 / gamma ] expr.
! => sqrt(pi)/2
```

6.7. Symmetry

$$\int_0^{\pi/4} \log(1 + \tan x) dx = \frac{\pi \log 2}{8}$$

Here $f(x) + f(\pi/4 - x)$ is constant, so the integral follows from that constant and the width of the interval, with no antiderivative.

```
symbolic[ 1 x tan + log ] symbolic[ x ] 0 symbolic[ pi 4 / ]
definite-by-symmetry expr.
! => log(2)*pi/8
```

6.8. Differentiating under the integral sign

$$F(t) = \int_0^1 (tx^2 + x) dx, \quad F'(t) = \int_0^1 x^2 dx$$

```
symbolic[ t x 2 ^ * x + ] symbolic[ x ] 0 1 symbolic[ t ]
feynman-derivative .
! => 1/3
symbolic[ t x 2 ^ * x + ] symbolic[ x ] 0 1 symbolic[ t ] 0
feynman-solve expr.
! => t/3 + 1/2
```

7. Limits and series

7.1. Indeterminate quotients

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

```
symbolic[ x sin x / ] symbolic[ x ] 0 limit .
! => 1
symbolic[ 1 x cos - x 2 ^ / ] symbolic[ x ] 0 limit .
! => 1/2
```

7.2. Zero times infinity

$$\lim_{x \rightarrow 0} x \log x = 0$$

```
symbolic[ x x log * ] symbolic[ x ] 0 limit .
! => 0
```

7.3. Limits at infinity

$$\lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^2 - 1} = 2, \quad \lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$$

```
symbolic[ 2 x 2 ^ * 3 + x 2 ^ 1 - / ] symbolic[ x ] symbolic[ inf ] limit .
! => 2
symbolic[ x x exp / ] symbolic[ x ] symbolic[ inf ] limit .
! => 0
symbolic[ x atan ] symbolic[ x ] symbolic[ inf ] limit expr.
! => pi/2
```

7.4. One to the infinity

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e, \quad \lim_{x \rightarrow 0} (1 + \sin 5x)^{7/x} = e^{35}$$

Writing f^g as $\exp(g \log f)$ turns this into a product the engine can already do.

```
symbolic[ 1 x + 1 x / ^ ] symbolic[ x ] 0 limit expr.
! => e
symbolic[ 1 5 x * sin + 7 x / ^ ] symbolic[ x ] 0 limit expr.
! => exp(35)
```

An exponent that tends to 0 is not indeterminate, and needs no such treatment.

```
symbolic[ 1 5 x * sin + x 7 / ^ ] symbolic[ x ] 0 limit .
! => 1
```

7.5. No limit

$\lim_{x \rightarrow \infty} \sin x$ does not exist

```
symbolic[ x sin ] symbolic[ x ] symbolic[ inf ] limit expr.
! => limit(sin(x), x, inf)
```

7.6. Maclaurin series

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$$

```
symbolic[ x exp ] symbolic[ x ] 3 maclaurin expr.
! => x^3/6 + x^2/2 + x + 1
symbolic[ x sin ] symbolic[ x ] 5 maclaurin expr.
! => x^5/120 - x^3/6 + x
symbolic[ 1 x + log ] symbolic[ x ] 3 maclaurin expr.
! => x^3/3 - x^2/2 + x
```

7.7. Expanding about another point

e^x about $x = 1$

```
symbolic[ x exp ] symbolic[ x ] 1 2 taylor expr.
! => e + e*(x - 1) + e*(x - 1)^2/2
symbolic[ x 3 ^ 2 x 2 ^ * - 5 + ] symbolic[ x ] 4 maclaurin expr.
! => x^3 - 2*x^2 + 5
```

8. Rules, compilation and numerics

8.1. The Pythagorean identity

$$\sin^2 x + \cos^2 x = 1$$

A rule set rewrites matching subexpressions, in any order of terms, and leaves the rest alone.

```
symbolic[ x sin 2 ^ x cos 2 ^ + ] pythagorean-rules rewrite .
! => 1
symbolic[ y 1 + sin 2 ^ y 1 + cos 2 ^ + z + ] pythagorean-rules rewrite expr.
! => z + 1
symbolic[ x sin 2 ^ y cos 2 ^ + ] pythagorean-rules rewrite expr.
! => cos(y)^2 + sin(x)^2
```

8.2. Rules you write

A rule literal uses pattern variables written ?x.

```
symbolic[ a b + c * ] rule[ ?x ?y + ?z * => ?x ?z * ?y ?z * + ] larray
rewrite expr.
! => a*c + b*c
symbolic[ x y 3 ^ * log ] log-expand-rules rewrite expr.
! => 3*log(y) + log(x)
symbolic[ x exp y exp * z * ] exp-rules rewrite expr.
! => z*exp(x + y)
```

8.3. Conditional rules

A rule can carry a test, here that the exponent is an integer.

```
symbolic[ x 2 ^ log x y ^ log + ]
symbolic[ ?a ?n ^ log ] symbolic[ ?n ?a log * ] [ "n" binding integer? ]
<conditional-rule> larray rewrite expr.
! => 2*log(x) + log(x^y)
```

8.4. Compiling an expression

An expression becomes a quotation, or a compiled word, taking the variables' values.

```
3 4 symbolic[ x 2 ^ y 2 ^ + ] { T{ sym f "x" } T{ sym f "y" } }
expr>quot call( x y -- r ) .
! => 25
2 symbolic[ x -3 ^ ] { T{ sym f "x" } } expr>quot call( x -- r ) .
! => 1/8
```

8.5. Numeric integration

$$\int_0^1 e^{-x^2} dx \approx 0.7468$$

```
symbolic[ x 2 ^ neg exp ] T{ sym f "x" } 0 1 nintegrate .
! => 0.7468241328202144
```

8.6. Checking an antiderivative numerically

Differentiating the result and evaluating both at a point is a quick check that a rule did the right thing.

```
symbolic[ x 2 ^ x sin * ] symbolic[ x ] integrate
symbolic[ x ] differentiate { { T{ sym f "x" } 0.7 } } subs evalf .
! => 0.31566666674646854
symbolic[ 0.7 2 ^ 0.7 sin * ] evalf .
! => 0.31566666674646854
```